

Smart Router

Track 1 — General-Purpose AI Agent

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AMD Developer Hackathon — ACT II
Team Robot ROCm

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`ghcr.io/wilfred-dore/track1-smart-router:v20`

The problem: accuracy is a gate, tokens are the score

Enterprise reality

- Not every task deserves a premium model
- Host cheap local models in-house, call the paid API only when it matters
- Track 1 scores exactly that: pass the 80% accuracy gate, then rank by **fewest Fireworks tokens**

Consequences for design

- Local inference = 0 tokens $\Rightarrow$ use it aggressively
- Every escalation must be *deliberate*
- Route by **verifiability**, not difficulty

8 task categories: factual QA · math · sentiment · summarization · NER · code debugging · logic · code generation

Architecture: a single-pass cascade, never a loop

1. **Deterministic solvers** — exact arithmetic parser, negation-aware sentiment lexicon, spaCy statistical NER with rule overrides; each self-demotes on doubt

0 tokens



2. **Local LLM** — Qwen2.5-3B, 4-bit GGUF, llama.cpp, CPU-only, bundled: summarization and code generation, verified for free (sentence counts, `ast.parse`)

0 tokens



3. **Zero-cost gate** — what cannot be *verified* for free escalates: math, logic, factual, code debugging

0 tokens



4. **Fireworks escalation** — model from `ALLOWED_MODELS`; `reasoning_effort=none`; logic solo (deterministic), the rest in two type-grouped batched calls

billed

Measured, not assumed: the descent from 6,559 to 639 tokens

Step	Change	Tokens	Accuracy
v11	everything escalated, 18 calls	6,559	94.7% (
+batch	escalations grouped: one thinking session	3,510	19/19 lo
+local	summarization & code answered on-device	2,758	19/19
+reasoning_effort=none	billed thinking switched off	548	19/19
+prompt trim, +solo logic	final profile (v20)	639–649	19/19

- Every step measured on the **real Fireworks API** with the real kimi-k2p7-code
- Batching suppresses billed reasoning (–74% measured in the literature; we observed the same)
- Batched logic without thinking is nondeterministically wrong → logic runs solo, ~6 tokens per task

Validated three ways before ever resubmitting

- **Real-API evaluation:** 19 practice-style fixtures, 19/19 on four consecutive runs, 639–649 tokens (near-deterministic)
- **Generalization gauntlet:** 19 fresh adversarial tasks written to break the pipeline — caught a sentiment-lexicon trap, a semantic code-debug miss and NER label errors; all fixed, 18/19 (the last miss is genuinely ambiguous)
- **Independent LLM-judge simulation:** a different model family graded all 38 answers the way the official judge would — caught nonsense justifications a regex checker cannot see
- **CI replays the grading environment exactly** (linux/amd64, 4 GB RAM, 2 vCPU): 19 tasks in 90 s, 1.97 GB compressed image, schema and size gates before any publish
- Everything config-driven; no hardcoded answers, no caching across runs; the only runtime network call is `FIREWORKS_BASE_URL`

Takeaways

- **Zero-cost first:** solvers + a bundled 3B absorb everything that can be verified for free; the paid API sees only the hard residue
- **Route by reliability, not difficulty:** categories where small models fail *silently* escalate outright
- **Measure everything:** stand-in proxies before credits, real-API sweeps after, an adversarial gauntlet and a simulated judge before each freeze
- **Plumbing wins hackathons:** most failed submissions break on infrastructure — ours is CI-proven against the exact grading limits

Submission

`ghcr.io/wilfred-dore/track1-smart-router:v20`

public · linux/amd64 · weights bundled · no secrets inside

Techniques adapted from publicly shared competitor write-ups and the literature are credited in the repository.